

# Fowler (2024) on the Japanese commuting zones

August 26, 2026

## What this deck reports

- ▶ Every table and figure of Fowler (2024) that the Japanese data supports, in the order that paper presents them.
- ▶ Delineations for 1980 to 2020; this deck shows the 2010 against 2020 comparison, the same decade spacing as the source paper.
- ▶ The wage-correlation block, that paper's Figure 2 and the Connection rows of its Table 2, waits on the Basic Survey on Wage Structure microdata.
- ▶ Every body slide carries a button to the appendix slide that shows all nine census years.

Source paper: <https://doi.org/10.1038/s41597-024-03829-5>

## The delineation

For municipalities  $i$  and  $j$  with commuting flows  $f_{ij}$  and resident labour forces  $W_i$ , the proportional flow is

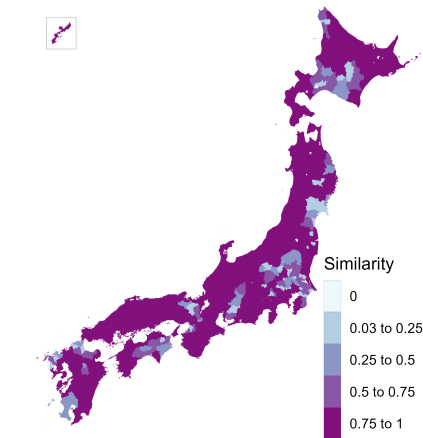
$$p_{ij} = \frac{f_{ij} + f_{ji}}{\min(W_i, W_j)}, \quad p_{ij} \leq 0.999,$$

and the dissimilarity is  $1 - p_{ij}$  with a zero diagonal.

- ▶ Average-linkage hierarchical clustering, cut at a fixed tree height of 0.977.
- ▶ Commuting matrix: residents mainly working, ordinary households, the only definition available for 2020.
- ▶ 1,656 municipalities: the four main islands and the Okinawa main island, plus every municipality joined to one by a permanent road link. The twenty-three special wards of Tokyo count as one, as the ordinance-designated cities do.
- ▶ Cores and urban areas for the fit statistics: the Urban Employment Area central cities and their areas.

# Similarity between the 2010 and the 2020 delineations

Figure 1 of Fowler (2024)



Jaccard similarity: the overlap of a municipality's two commuting zones over their union. Mean 0.85.

# Diagnostic statistics

Table 1 of Fowler (2024)

|                                      | 2010      | 2020      |
|--------------------------------------|-----------|-----------|
| Municipalities in the delineation    | 1,656     | 1,655     |
| Commuting zones                      | 231       | 223       |
| Single-municipality zones            | 9         | 10        |
| Municipalities in the largest zone   | 27        | 29        |
| Non-contiguous zones                 | 0         | 1         |
| Smallest zone, resident labour force | 530       | 475       |
| Average zone, resident labour force  | 202,694   | 207,532   |
| Largest zone, resident labour force  | 4,929,463 | 5,624,897 |
| Smallest zone area (sq. km)          | 184       | 100       |
| Average zone area (sq. km)           | 1,574     | 1,630     |
| Largest zone area (sq. km)           | 5,744     | 5,268     |
| Compactness of zones                 | 0.30      | 0.30      |

Compactness is the Polsby–Popper ratio,  $4\pi A/P^2$  of the dissolved zone, averaged across zones.

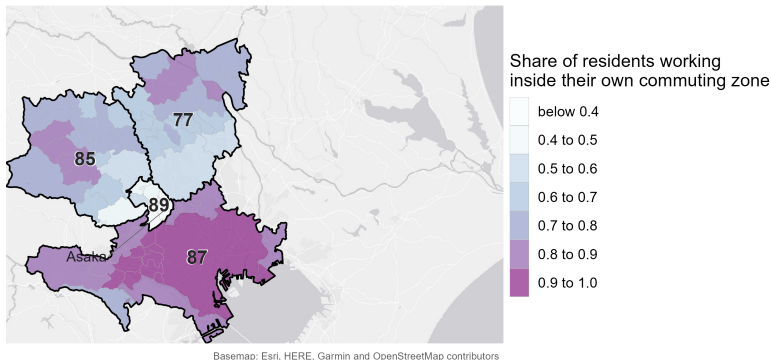
# Fit statistics

Table 2 of Fowler (2024)

|                                                 | 2010    | 2020  |
|-------------------------------------------------|---------|-------|
| <i>Core</i>                                     |         |       |
| Urban areas split across zones                  | 31      | 33    |
| Zones containing a core                         | 72.7%   | 67.3% |
| Residents working in a core of their zone       | 52.3%   | 51.9% |
| Workforce living in a core of their zone        | 47.9%   | 47.7% |
| <i>Connection</i>                               |         |       |
| Minimum wage correlation                        | pending |       |
| Mean wage correlation                           | pending |       |
| Mean wage correlation, multi-municipality zones | pending |       |
| <i>Containment</i>                              |         |       |
| Minimum contained                               | 44.5%   | 40.5% |
| Mean contained                                  | 90.0%   | 90.4% |
| Share of the labour force contained             | 85.8%   | 86.1% |

# The commuting zone with the lowest mean containment

Figure 3 of Fowler (2024)



Six municipalities in southern Saitama at 0.41, between the Tokyo zone and the zones beyond.

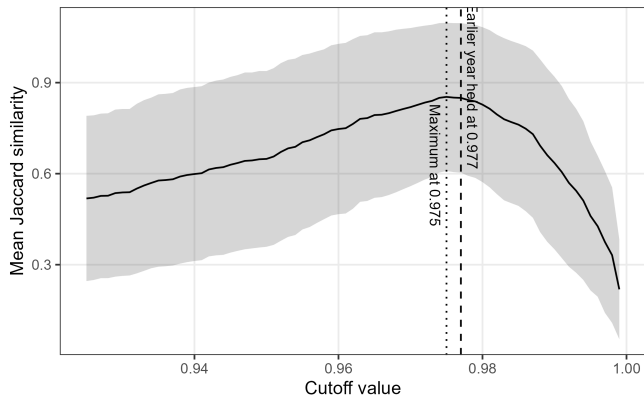
► The same map at 0.980

## Technical validation

- ▶ The cutoff. It is inherited rather than derived. Does the similarity between delineations pick out a better value?
- ▶ Non-contiguity. Zones that hold a municipality sharing its zone with none of its neighbours.
- ▶ Split urban areas. Externally defined urban areas cut across more than one commuting zone.

# Similarity as a function of the cutoff

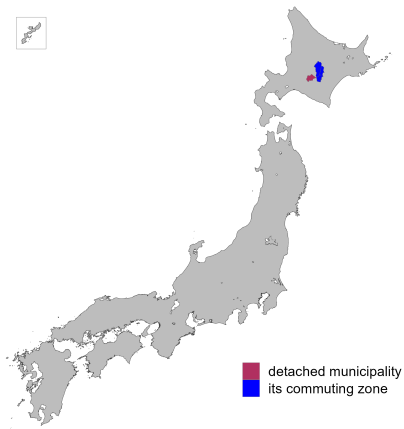
Figure 4 of Fowler (2024)



Holding 2010 at 0.977, the peak is 0.85 at 0.975, and the curve stays within 0.005 of it from 0.974 to 0.977, spanning 236 down to 223 zones.

# Non-contiguous municipalities

Figure 5 of Fowler (2024)



One municipality in 2020, Shimukappu in Hokkaido, 991 metres from the nearest part of its own zone.

# Reassignments that would remove non-contiguity

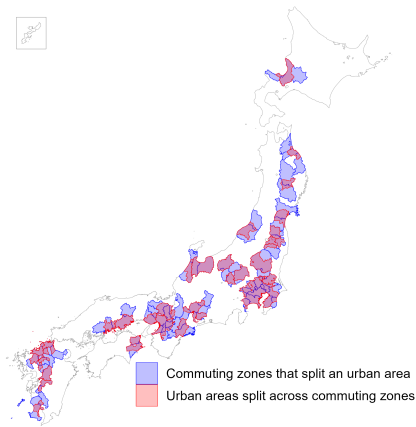
Table 3 of Fowler (2024)

| Municipality | Zone | To own zone | Nearest zone | To that zone | Suggestion |
|--------------|------|-------------|--------------|--------------|------------|
| Shimukappu   | 26   | 95.9%       | 18           | 2.9%         | Retain     |

Its own zone takes 96 percent of its commuters, the adjacent zone 3 percent.

# Urban areas split across commuting zones

Figure 6 of Fowler (2024)



33 of the 208 Urban Employment Areas in 2020, concentrated along the Pacific corridor.

# Appendix

## Diagnostic statistics, 1980 to 2020

| Year | Municipalities | Commuting zones | Single-municipality | Largest zone | Non-contiguous |
|------|----------------|-----------------|---------------------|--------------|----------------|
| 1980 | 1,656          | 384             | 71                  | 31           | 2              |
| 1985 | 1,656          | 341             | 53                  | 29           | 2              |
| 1990 | 1,656          | 307             | 34                  | 28           | 1              |
| 1995 | 1,656          | 281             | 24                  | 28           | 2              |
| 2000 | 1,656          | 261             | 15                  | 30           | 1              |
| 2005 | 1,656          | 239             | 12                  | 27           | 0              |
| 2010 | 1,656          | 231             | 9                   | 27           | 0              |
| 2015 | 1,651          | 225             | 9                   | 29           | 2              |
| 2020 | 1,655          | 223             | 10                  | 29           | 1              |

Counts at the 0.977 cutoff. Non-contiguous zones are those holding a municipality with no neighbour in its own zone.

## Zone size and shape, 1980 to 2020

| Year | Resident labour force |         |           | Area (sq. km) |         |         | Compactness |
|------|-----------------------|---------|-----------|---------------|---------|---------|-------------|
|      | Smallest              | Average | Largest   | Smallest      | Average | Largest |             |
| 1980 | 286                   | 120,319 | 4,747,659 | 16            | 947     | 3,464   | 0.33        |
| 1985 | 650                   | 140,701 | 5,265,234 | 24            | 1,066   | 3,839   | 0.32        |
| 1990 | 788                   | 168,711 | 5,672,523 | 24            | 1,184   | 4,429   | 0.32        |
| 1995 | 723                   | 189,749 | 5,312,186 | 24            | 1,294   | 4,429   | 0.32        |
| 2000 | 523                   | 202,706 | 6,000,065 | 109           | 1,393   | 4,429   | 0.31        |
| 2005 | 295                   | 211,525 | 5,355,045 | 109           | 1,521   | 4,733   | 0.31        |
| 2010 | 530                   | 202,694 | 4,929,463 | 184           | 1,574   | 5,744   | 0.30        |
| 2015 | 479                   | 205,844 | 4,819,056 | 100           | 1,613   | 4,909   | 0.30        |
| 2020 | 475                   | 207,532 | 5,624,897 | 100           | 1,630   | 5,268   | 0.30        |

Population is the resident labour force carried in the commuting matrix. Compactness is the Polsby–Popper ratio.

## Containment, 1980 to 2020

| Year | Minimum | Mean  | Labour-force weighted mean | Share of the labour force | Mean work contained |
|------|---------|-------|----------------------------|---------------------------|---------------------|
| 1980 | 43.6%   | 91.1% | 88.7%                      | 88.7%                     | 94.5%               |
| 1985 | 23.0%   | 90.4% | 88.3%                      | 88.1%                     | 94.0%               |
| 1990 | 21.2%   | 89.9% | 87.6%                      | 87.1%                     | 93.4%               |
| 1995 | 20.9%   | 89.2% | 86.7%                      | 86.3%                     | 92.6%               |
| 2000 | 40.6%   | 89.9% | 87.0%                      | 86.6%                     | 92.3%               |
| 2005 | 44.7%   | 90.1% | 86.7%                      | 86.1%                     | 92.3%               |
| 2010 | 44.5%   | 90.0% | 86.5%                      | 85.8%                     | 92.1%               |
| 2015 | 39.0%   | 90.1% | 86.1%                      | 85.3%                     | 91.5%               |
| 2020 | 40.5%   | 90.4% | 87.0%                      | 86.1%                     | 91.4%               |

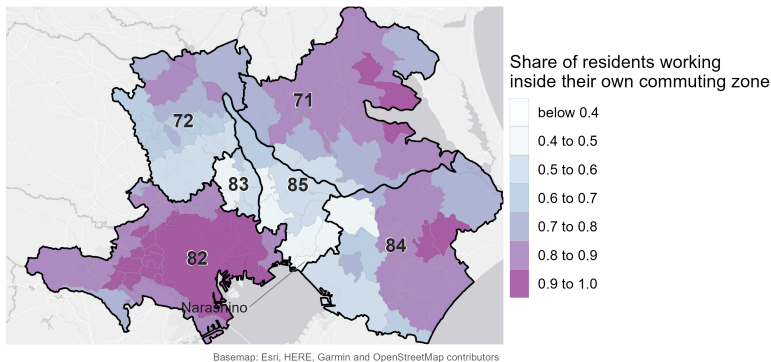
The minimum and the mean are taken over commuting zones; the share of the labour force is a single national ratio.

## Core measures, 1980 to 2020

| Year | Urban Employment Area |         |         |       | District population 10,000 or more |         |         |
|------|-----------------------|---------|---------|-------|------------------------------------|---------|---------|
|      | With core             | Work in | Live in | Split | With core                          | Work in | Live in |
| 1980 | 55.7%                 | 58.1%   | 53.5%   | 50    | 60.7%                              | 74.0%   | 72.5%   |
| 1985 | —                     | —       | —       | —     | 64.5%                              | 74.0%   | 72.7%   |
| 1990 | 61.6%                 | 55.8%   | 51.2%   | 46    | 68.1%                              | 74.4%   | 73.2%   |
| 1995 | 65.5%                 | 54.5%   | 49.9%   | 50    | 71.9%                              | 74.4%   | 73.4%   |
| 2000 | 68.6%                 | 54.3%   | 49.8%   | 55    | 74.7%                              | 74.8%   | 73.9%   |
| 2005 | 71.1%                 | 52.8%   | 48.5%   | 39    | 77.0%                              | 74.4%   | 73.6%   |
| 2010 | 72.7%                 | 52.3%   | 47.9%   | 31    | 77.9%                              | 74.0%   | 73.2%   |
| 2015 | 73.3%                 | 51.1%   | 47.0%   | 34    | 78.2%                              | 73.3%   | 72.7%   |
| 2020 | 67.3%                 | 51.9%   | 47.7%   | 33    | 73.1%                              | 74.4%   | 73.9%   |

The district rule marks a municipality as a core on its densely inhabited district population alone, using no commuting data. There is no 1985 Urban Employment Area delineation.

## The lowest mean containment at the 0.980 cutoff



Same extent as the previous map. Zone 89 is absorbed into Tokyo, which becomes zone 82; the minimum moves to Chiba, at 0.50.

## The two cutoff anchors compared

| Year | Cutoff | Commuting zones | Non-contiguous | Minimum contained | Mean contained | Share of the labour force | Areas split |
|------|--------|-----------------|----------------|-------------------|----------------|---------------------------|-------------|
| 2010 | 0.977  | 231             | 0              | 44.5%             | 90.0%          | 85.8%                     | 31          |
| 2020 | 0.977  | 223             | 1              | 40.5%             | 90.4%          | 86.1%                     | 33          |
| 2010 | 0.980  | 216             | 0              | 48.7%             | 90.8%          | 86.3%                     | 28          |
| 2020 | 0.980  | 208             | 1              | 49.9%             | 90.8%          | 86.5%                     | 32          |

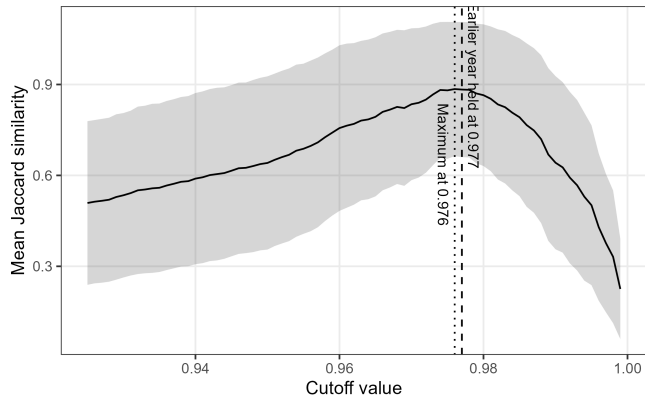
Moving from 0.977 to 0.980 removes fifteen commuting zones in each year and lifts the minimum containment by four to nine points.

## Every comparison pair

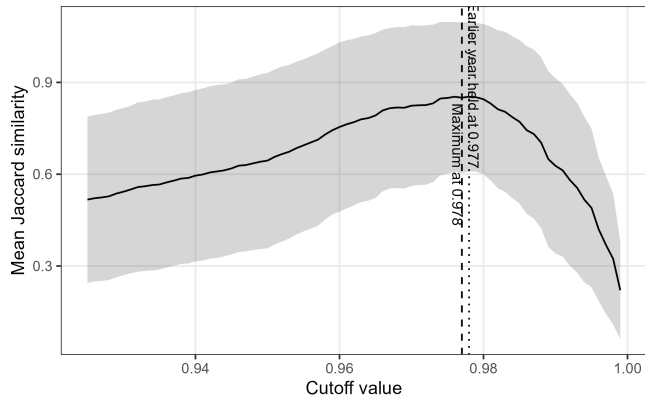
| Pair              | Role     | Municipalities | Mean | Labour-force weighted |
|-------------------|----------|----------------|------|-----------------------|
| 1980 against 1990 | appendix | 1,656          | 0.77 | 0.75                  |
| 1990 against 2000 | appendix | 1,656          | 0.83 | 0.86                  |
| 2000 against 2010 | appendix | 1,656          | 0.85 | 0.86                  |
| 2005 against 2015 | baseline | 1,651          | 0.85 | 0.83                  |
| 2010 against 2015 | baseline | 1,651          | 0.88 | 0.87                  |
| 2010 against 2020 | baseline | 1,655          | 0.85 | 0.82                  |

The 2010 to 2020 comparison sits below both pre-pandemic controls but above the decade pairs of the 1980s and 1990s.

## Similarity against the cutoff, 2010 against 2015



## Similarity against the cutoff, 2005 against 2015



## One departure from the existing pipeline

The existing Japanese pipeline joins the flow file to its own transpose in a way that keeps a pair only when the flow from  $i$  to  $j$  is recorded, so a pair connected only in the reverse direction enters as unconnected.

- ▶ The matrix it clusters is asymmetric; this replication uses the symmetric numerator the source method specifies.
- ▶ Between 1.2 and 4.4 percent of pairs are affected in each year, moving up to four zones and up to 108 municipalities.
- ▶ Measured year by year in `validation/notes/method_crosscheck.md`.

## Where everything lives

On the repo: [https://github.com/daisukeadachi/commuting\\_zone\\_japan/](https://github.com/daisukeadachi/commuting_zone_japan/)

- ▶ `validation/notes/replication_results.md` — the written results, with every number in this deck in context.
- ▶ `validation/notes/method_crosscheck.md` — the delineation formula across four implementations.
- ▶ `validation/notes/core_definition.md` — why the Urban Employment Area supplies the cores.
- ▶ `validation/output` — the tables behind every slide, as comma-separated files.
- ▶ `validation/code` — the nine scripts, run in the order they are listed in the results note.